

Numerical study on the punching shear strength of edge steel-reinforced concrete slab-column connections

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ABSTRACT

Punching shear failure in steel-reinforced concrete (RC) edge slab-column connections poses a critical challenge in structural engineering, with safety and design efficiency implications. This study investigates the influence of key geometric parameters—slab thickness, column aspect ratio, square column stub size, and span-to-depth ratio—on punching shear strength, a topic underexplored in existing research. A unique feature of this research is the ability to examine the effects of slab thickness, column aspect ratio, and square column stub size independently, without any interference from changes in the span-to-depth ratio, by keeping it constant. This approach has never been achieved before and is impractical in experimental studies. Using finite element modeling, over 20 connection configurations were analyzed to assess their structural behavior. Results showed that increasing slab thickness reduced deflection by up to 95 % but decreased punching shear strength by 40 % due to stress redistribution. Higher column aspect ratios and larger square column stubs caused strength reductions of 36 % and 43 %, respectively, while reducing the span-to-depth ratio enhanced stiffness and punching shear strength by 25 %. The study also evaluated the accuracy of three design standards—American (ACI 318–19 (22)), Canadian (CSA-A23.3:24), and Japanese (JSCE-2007)—revealing deviations of up to 144 % from actual performance. To address these discrepancies, four new equations were proposed, tailored to specific geometric parameters, and validated against available literature, demonstrating superior accuracy compared to existing standards. These findings underscore the limitations of current methodologies and emphasize the importance of incorporating geometric factors to improve slab-column connection designs in modern construction.

1. Introduction

Steel-reinforced concrete (RC) edge slab-column connections are essential components in many structures. These connections often experience high stress concentrations, making them susceptible to complex punching shear failures, which highlights the importance of understanding their behavior under varying loads. Despite their significance, these connections have received little attention. Researchers have examined factors such as reinforcement layout and shear reinforcement to assess their effects on connection performance.

Early studies showed that concentrated forces near column perimeters could cause brittle punching shear failures, emphasizing the importance of reinforcement for improving load redistribution and resilience [1]. Subsequent investigations highlighted the role of reinforcement ratios in shear capacity, especially in edge connections where asymmetric support conditions lead to unique stress distributions [2].

Further research on edge slab-column connections has focused on

challenges such as load eccentricity and inadequate confinement of the edge slab. One proposed solution is the use of headed studs to improve punching resistance and minimize deflection [3]. Experimental studies have shown that shear reinforcement significantly enhances punching shear capacity, shifting failure modes from brittle to ductile and thereby improving safety margins [4]. Additionally, the design of shear reinforcement considers factors such as slab flexibility, moment transfer, and load eccentricity, which are especially crucial in high-rise structures subjected to significant eccentric loads [5].

These findings collectively highlight the critical need for specialized reinforcement strategies, particularly for steel-RC edge slab-column connections. This underscores the importance of developing advanced design guidelines to address the growing demands of modern structures. As buildings are subjected to higher loads, implementing tailored reinforcement solutions becomes essential for ensuring both strength and ductility. Such strategies play a key role in the safe and resilient design of contemporary buildings, enabling them to withstand dynamic and

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increasing load demands.

While these studies have advanced the understanding of steel RC slab-column connections, they have not comprehensively addressed several important geometric parameters influencing structural performance. Critical factors such as column aspect ratio, span-to-depth ratio, slab thickness, and perimeter-to-depth ratio have received limited attention in existing literature. These parameters can play a pivotal role in determining punching shear capacity and failure mechanisms, especially in edge connections where asymmetrical support conditions lead to distinctive stress distributions. The current article aims to address these gaps by specifically investigating the effects of these geometric factors on slab-column connection performance.

Given the practical challenges of altering slab and column dimensions in a laboratory environment and the necessity of adjusting the test setup for each prototype, conducting a comprehensive experimental study on these variables proved highly difficult, if not impossible [6–7]. As a result, the study utilized a finite element (FE) modeling approach to simulate a wide range of structural configurations. The FE program enabled a more flexible and detailed analysis, providing insights that would be difficult to obtain through physical experiments alone [8–10]. This modeling approach allows for a broader investigation into the impact of geometric factors on the punching shear behavior of steel RC edge slab-column connections. However, an experimental study is still needed to verify the model's accuracy and ensure its predictions align with real-world performance. Such experimental validation would also

help refine the model, enhancing its reliability and practical application [11–12].

2. Previously examined edge slabs

Four full-scale steel-RC edge slab-column joints documented in the literature [13] were selected for this study. These connections were specifically chosen to assess the accuracy of the finite element (FE) model. The following sections summarize and provide key details of these four connections.

2.1. General layout

The slabs had a depth of 120 mm, with in-plane dimensions of 1540 mm by 1020 mm. The columns featured square cross-sections, each measuring 250 mm, and the height of the columns extending above and below the slab was 700 mm (Fig. 1). The slabs were cast with tensile reinforcement positioned at the bottom and compressive reinforcement at the top. The average tensile reinforcement ratio, both parallel and perpendicular to the free edge, was 0.0075, using No 10 M bars. The compressive reinforcement ratio was 0.0045 in both directions, utilizing No 5 M bars (Table 1). In the compressive region, two No 10 M bars were inserted, extending through the column in both directions to ensure structural integrity. The columns were over-reinforced with No 25 M bars, and 8 mm-diameter plain bars were used as ties to prevent

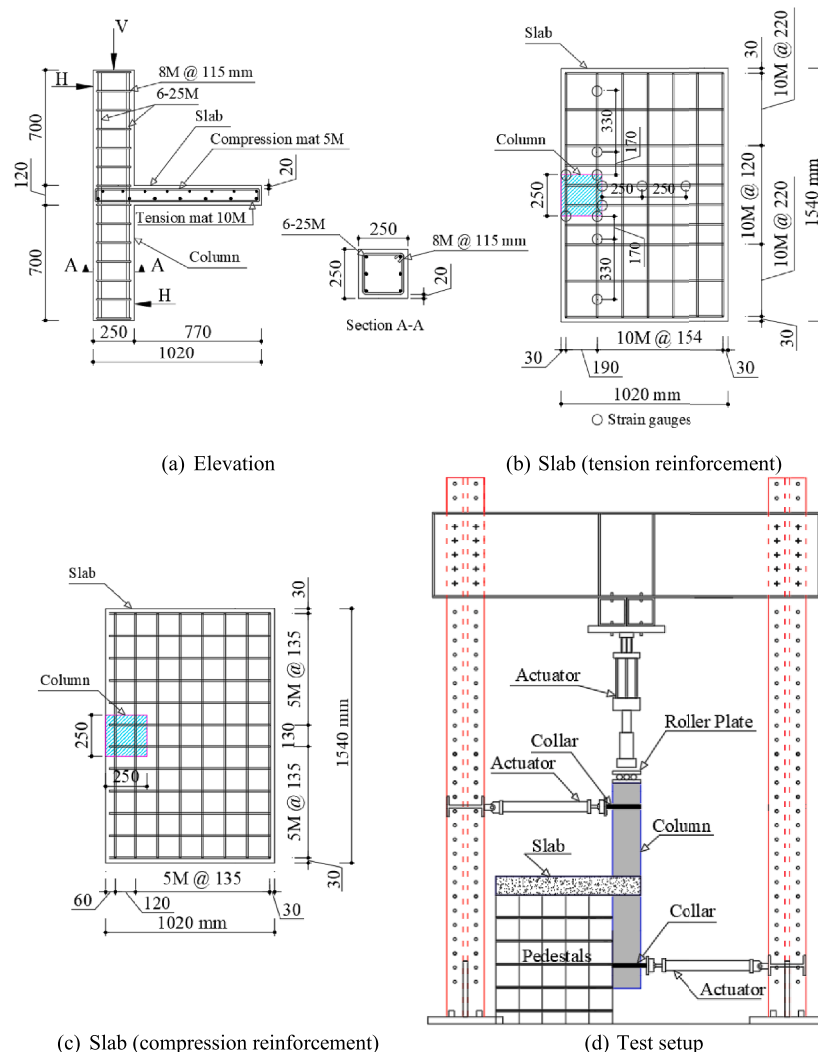


Fig. 1. Specimen XXX configuration and test setup (recreated from El-Salakawy et al. 1999).

Table 1
Rebars' properties.

Type	Diameter (mm)	Area (mm ²)	Yield strength (MPa)	Modulus of elasticity (GPa)
10M	11.3	100	545	195
5M	7.0	38	430	189

premature failure. Fig. 1 provides the dimensions and reinforcement details of the prototypes.

Each slab is identified by a three-character title. "XXX" denotes a slab without any opening. "SF0" refers to a slab with a 150 × 150 mm opening placed in front of the column, aligned with the direction of the unbalanced moment. "SE0" indicates a slab with a 150 × 150 mm opening positioned at the side of the column, perpendicular to the direction of the unbalanced moment. "SF1" describes a slab with a 150 × 150 mm opening similar to SF0 but placed 90 mm (the effective depth) away from the column face.

2.2. Test setup

The test setup included a steel frame, I-beams, hydraulic actuators, transducers, and a data acquisition system. The vertical actuator applied an axial load to the upper column stub using a steel plate to evenly distribute the load, while two horizontal actuators applied equal and opposite horizontal forces near the column ends to create an unbalanced moment. The slabs were tested upside down to simulate tension cracks at the bottom and were supported on I-beams with steel and neoprene plates to ensure uniform reactions. The moment-to-shear ratio was maintained at 0.3 m throughout. The load was applied in three stages: Stage 1 applied load at 2.5 kN/min until the service load; Stage 2 cycled shear and moment 10 times between dead and live loads; and Stage 3 incrementally increased loads until failure. A diagram of the test setup is shown in Fig. 1.

2.3. Key observations

The shear strength of the connection is more significantly reduced by an opening at the front of the column (SF0) than by one located at the side (SE0). The opening near the front column face (SF0) reduced the shear strength by 10 %, while the side-facing opening (SE0) decreased the capacity by only 4 %. The position of the opening relative to the column also affected the connection's strength. In the specimen where the opening was positioned at a distance equal to the effective depth (d)

from the front column face (SF1), the shear strength was reduced by 8 %.

3. Numerical program

The four steel-RC edge slab-column connections selected from the literature [13] were simulated using ATENA 3D FE program [14]. A parametric study was then conducted to evaluate additional factors and enhance the understanding of this type of connection. The following sections summarize the properties and materials used to achieve this objective.

3.1. Components and materials used

In the current article, concrete behavior was modeled using a non-linear fracture framework known as CC3D-nonlincementitious2. The fracture properties were characterized using Rankine's criteria, while the Menétrey-Willam 3D failure model governed the non-linear behavior [14]. This framework was designed to replicate the cracking behavior of concrete under load. Fig. 2 demonstrated the stress-strain relationship as defined by this framework. Before reaching the failure point, the compressive properties of the concrete were determined based on the CEB-FIP Model Code provisions [15]. Beyond that point, the strength diminished progressively until complete degradation. In tension, concrete was assumed to behave linearly up to its tensile strength. Once failure occurred, the post-cracking behavior adhered to Hordijk's empirical crack-opening relationship [16].

The stiffness contribution of the reinforcing steel was incorporated into the overall stiffness matrix of the concrete. The mechanical properties of the steel bars, detailed in Table 1, were used to define their behavior. The stress-strain relationship for the steel bars was assumed to be linear, with a slope corresponding to the modulus of elasticity up to the yield point, followed by a horizontal plateau. The bond strength between the steel reinforcement and the surrounding concrete was accounted for in the analysis. This bond strength was determined according to the guidelines provided in the CEB-FIP Model Code [15], as depicted in Fig. 3.

3.2. Fabrication and verification of the connections

Four full-scale slabs, derived from the literature [13], were constructed using the material properties and specifications provided in Tables 1 and 2. Representative images of the standard prototype are shown in Fig. 4.

Shear forces were applied uniformly across the entire cross-section of

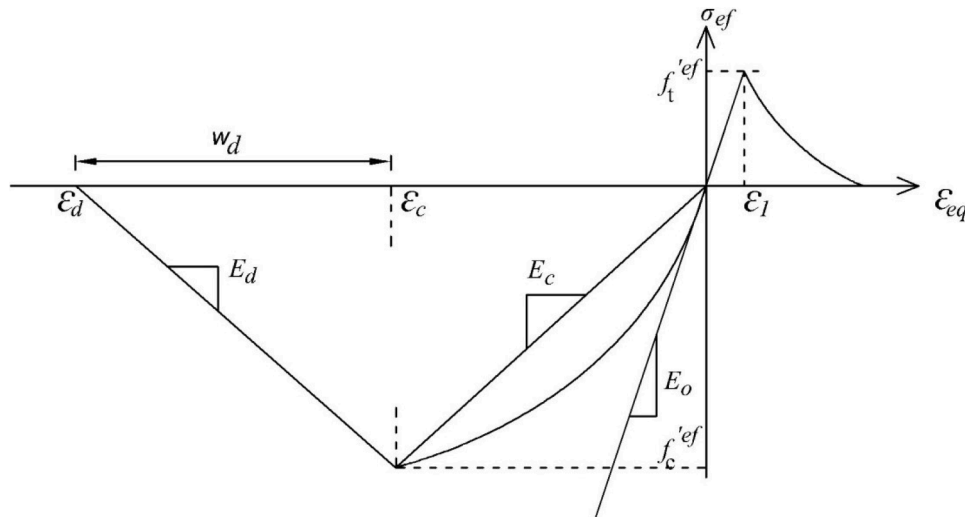


Fig. 2. Stress-strain for concrete (reproduced from Cervenka et al. 2013).

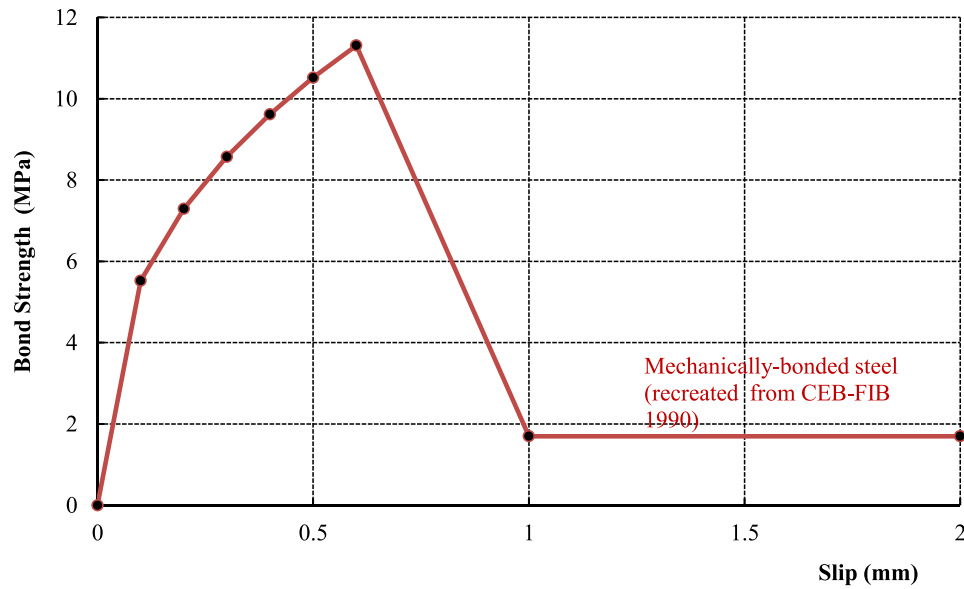


Fig. 3. Bond strength-slip relationship.

Table 2
Details.

Slab ID	f_c (MPa)	Size of the opening (mm)	Location of the opening from the column-face	Opening's distance from the column-face (mm)	Experimental ultimate load V_{exp} (kN)	FE ultimate load V_{mod} (kN)	V_{exp}/V_{mod}
XXX	33.0	N/A	N/A	N/A	125	126	0.99
SF0	31.5	150 × 150	Front	0.00	110	108	1.02
SE0	32.5	150 × 150	Edge	0.00	120	118	1.02
SF1	33.0	150 × 150	Front	90.0	115	120	0.96
Average							0.9975
S.D (%)							2.50
COV (%)							2.50

the upper column at a constant load of 1 kN, directed downward, replicating the experimental loading procedure. Simultaneously, an unbalanced moment was introduced at the top and bottom of the columns using horizontal forces calibrated to create a moment-to-vertical-force ratio of 0.3 m, ensuring alignment with the experimental test setup.

The simulation followed the three-step loading protocol employed in the experimental program to ensure consistency. In Stage 1, loads were applied up to the service limit. Stage 2 involved cycling the shear force and moment 10 times between dead and live load conditions. Finally, in Stage 3, loads were progressively increased until structural failure occurred. Newton's iterative method was utilized to compute the slabs' shear strength numerically, providing a robust approach to solving the non-linear equations governing the system.

Fig. 5 illustrates the crack patterns observed in both the experimental and numerical prototypes. The numerical simulation successfully replicated the experimental crack patterns with exceptional accuracy, demonstrating a high degree of reliability in capturing the failure mechanisms of the slabs. This close agreement highlighted the robustness of the modeling approach in predicting crack initiation and propagation under the applied loading conditions.

Fig. 6 presents the load-deflection relationship for both experimental and numerical models. The numerical results closely followed the experimental trend, showcasing consistent curvature and deflection behavior throughout the loading stages. This strong correlation underscored the validity of the numerical model in reproducing the structural response, particularly in simulating the stiffness and deformational characteristics of the joints under various load intensities.

Similarly, Fig. 7 compares the load-strain relationship in the reinforcing bars between the experimental and numerical prototypes. The numerical model demonstrated remarkable alignment with the experimental data, accurately capturing the strain variations in the reinforcement as the load increased.

Table 2 presents the shear strength values obtained from laboratory tests alongside those predicted through numerical analysis. The discrepancy between the two results was minimal, amounting to 4 % or less. The ratio of experimental shear strength to numerically predicted shear strength was calculated as 1.0 ± 0.03 , with a coefficient of variation (COV) of 2.5 %. These findings demonstrated the high accuracy of the numerical model in predicting the shear strength of slab-column connections.

The close alignment between the experimental and numerical outcomes confirmed the software's reliability and effectiveness in capturing the structural behavior of these joints under shear loading.

3.3. Parametric study

The developed model was utilized to investigate the influence of four key parameters on the structural performance of edge steel-RC slab-column connections:

1. Slab thickness: Varied between 150 mm and 350 mm in increments of 50 mm.
2. Column aspect ratio: Adjusted from 1.0 to 5.0 with an increment of 1.0.

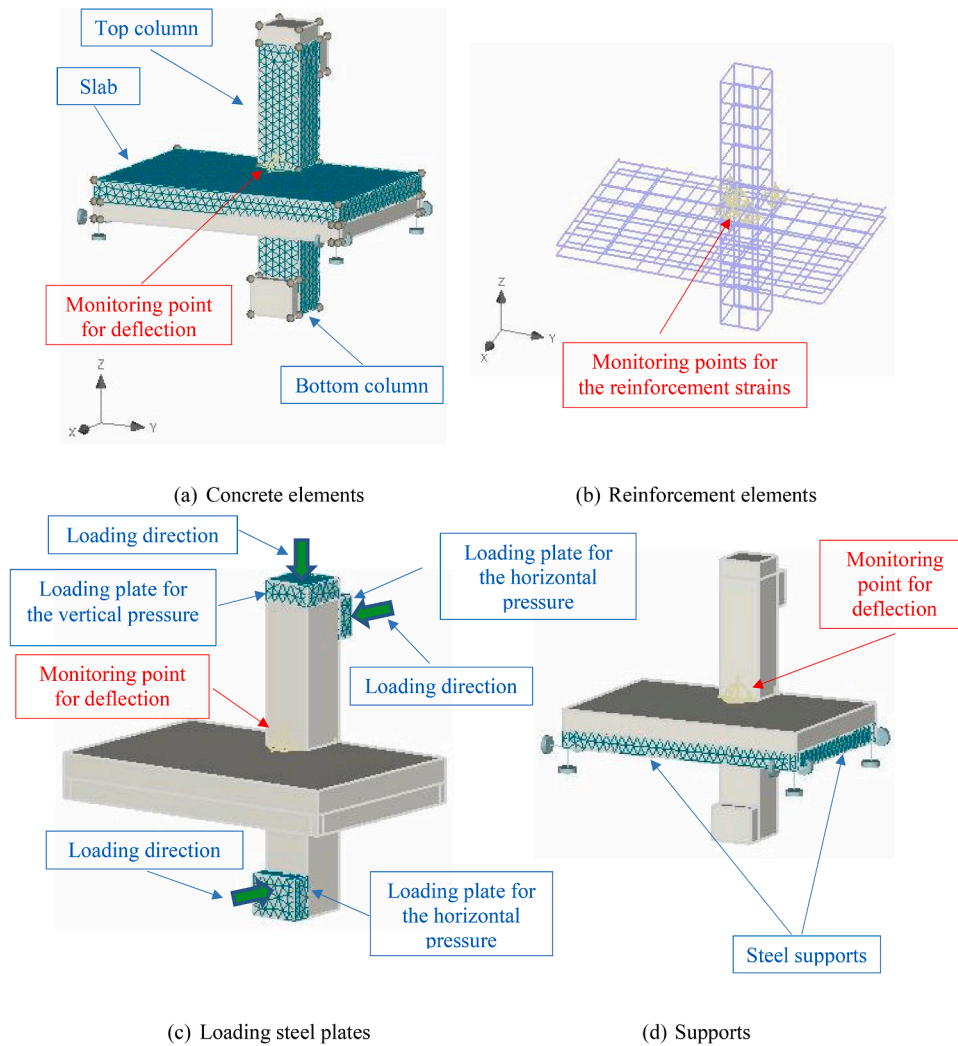


Fig. 4. The model configuration.

3. Column stub size: Square stubs ranging from 250 mm to 650 mm, increased in steps of 100 mm.
4. Span-to-depth ratio: Examined between 2.0 and 8.0 with an increment of 1.0.

A total of 20 connections were modeled to provide a comprehensive analysis of the selected parameters. Notably, for the first three parameters, the span-to-depth ratio was deliberately kept constant across all variations. This approach ensured that the results were not influenced by changes in the span-to-depth ratio, allowing for an isolated assessment of the targeted parameters. Maintaining this consistency required adjustments to the geometry of each slab and precise modifications to the support locations, ensuring accurate and unbiased comparisons throughout the analysis.

Such precise adjustments would be nearly impossible to replicate in a laboratory setting due to the extensive reconfiguration of formwork and test setup required for each connection. This practical limitation underscored the decision to conduct the research using FE analysis software. The FE program provided the flexibility to model, modify, and analyze each connection efficiently, enabling a detailed parametric study that would otherwise be unfeasible in a physical laboratory environment.

Each connection was labeled using a four-number code, where the first number represented the slab thickness in millimeters, the second indicated the column aspect ratio, the third corresponded to the smaller

dimension of the column cross-section in millimeters, and the fourth denoted the clear-span-to-effective-depth ratio. This concise system ensured clear identification and simplified reference during the analysis.

• Parameter 1: Slab thickness

To assess the impact of slab thickness, six connections were modeled. The first connection featured a thickness of 120 mm, replicating the dimensions used in the experimental study. The remaining connections were designed with an initial thickness of 150 mm, increasing incrementally by 50 mm up to 350 mm.

Fig. 8(a) illustrates the load-deflection curves for the tested connections with varying slab thicknesses. The load-deflection behavior demonstrated distinct phases reflecting the structural response under increasing loads. Initially, the connections exhibited linear elastic behavior, where deflection increased proportionally with the applied load. As the load continued to rise, cracking developed around the critical perimeter near the column, leading to a nonlinear response characterized by an increased deflection rate. In the final stage, the slabs approached the ultimate capacity, and the load-deflection curve flattened as significant deformation led to failure. This progression highlighted the combined influence of flexural stiffness, cracking resistance, and punching shear strength on the overall response.

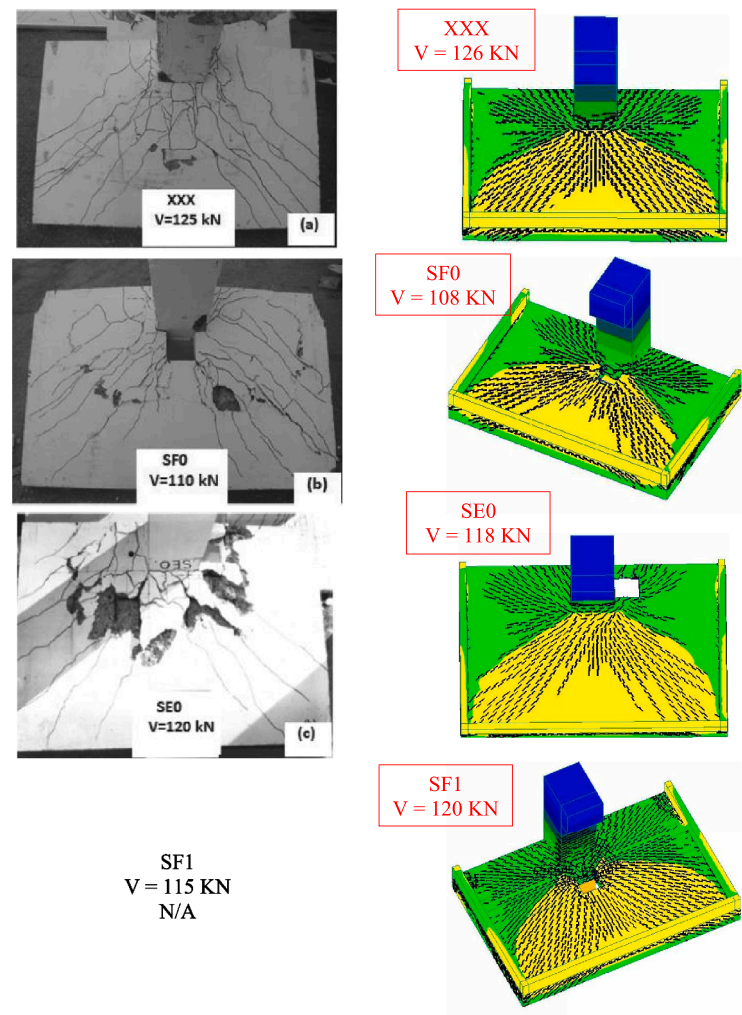


Fig. 5. Cracks at the (bottom) tension side of the slabs.

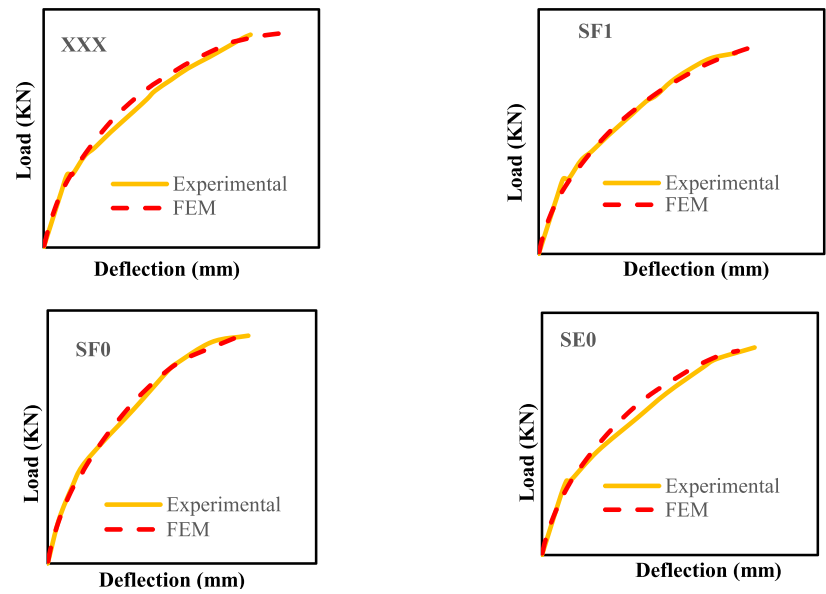


Fig. 6. Load-deflection curves.

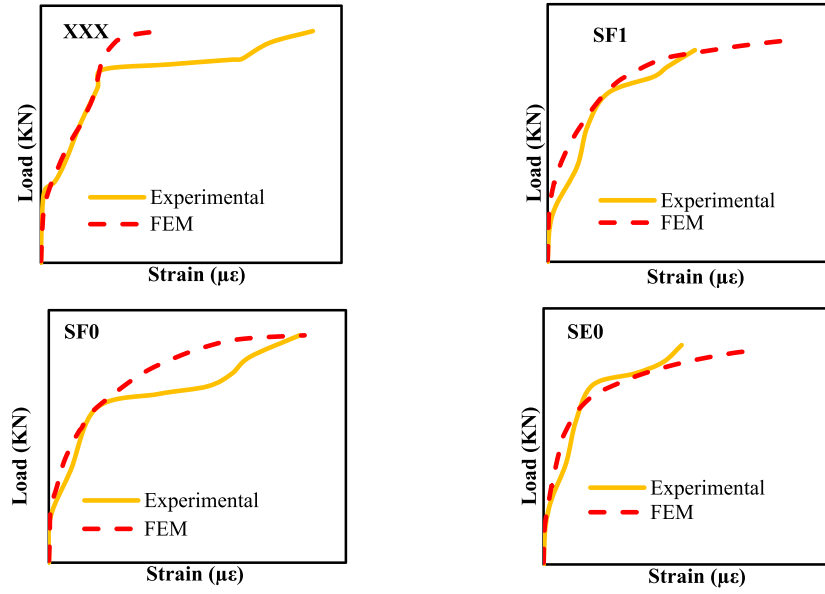


Fig. 7. Load-reinforcement strain curves.

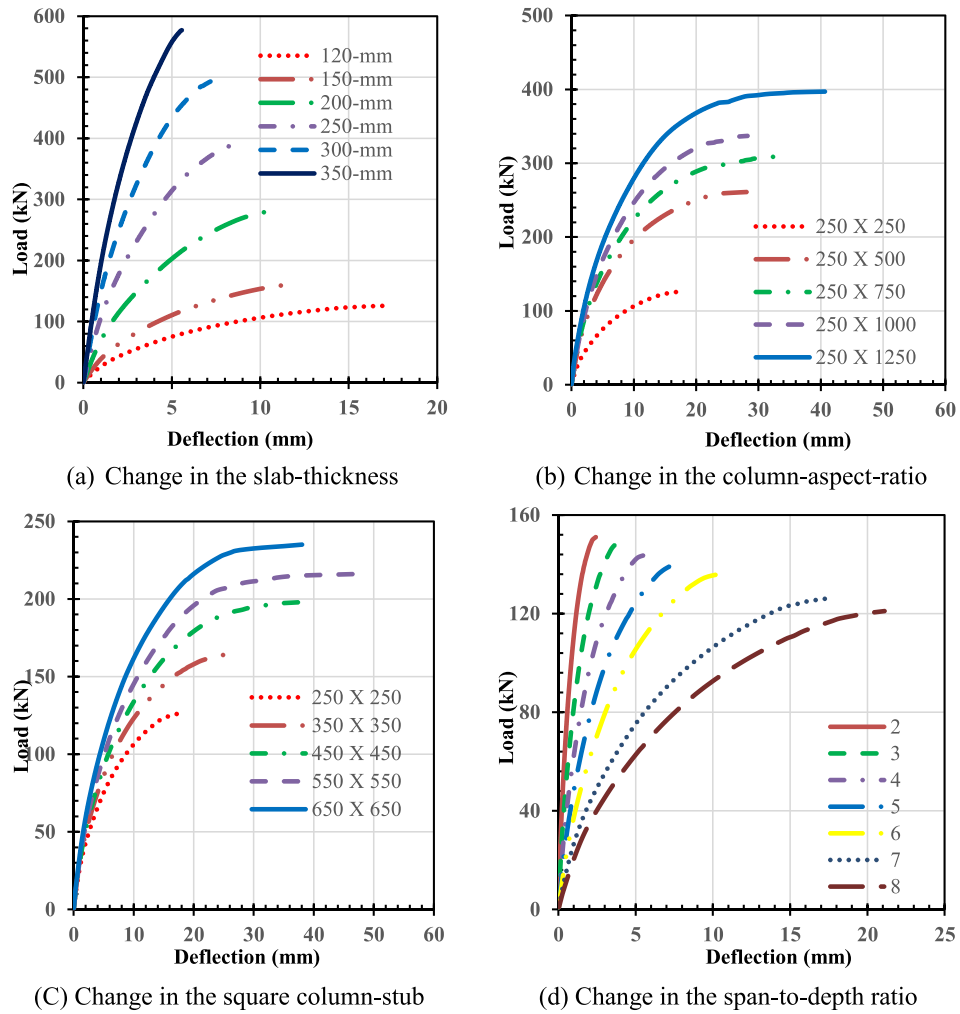


Fig. 8. Load-deflection curves for the studied parameters.

Increasing the slab thickness substantially reduced deflection. For slabs with thicknesses of 150 mm, 200 mm, 250 mm, 300 mm, and 350 mm, deflection reductions were 52 %, 81 %, 89 %, 93 %, and 95 %, respectively, compared to the 120-mm thick slab at the same shear level. At failure, the deflection reductions were less pronounced at 30 %, 41 %, 52 %, 58 %, and 68 %, respectively. These results indicate that while thicker slabs significantly enhance stiffness and reduce deformation under service conditions, the relative benefit diminishes as the slabs approach the ultimate capacity.

The reduction in deflection with increasing slab thickness can be attributed to higher flexural stiffness and improved shear resistance provided by thicker sections. Thicker slabs resist cracking more effectively, delaying the transition to nonlinear behavior and minimizing deflection throughout loading. Furthermore, they distribute stresses more efficiently, enhance resistance to punching shear failure and increase the load-bearing capacity. These observations underline the critical role of slab thickness in improving the structural performance of edge slab-column connections under both service and ultimate loading conditions.

Fig. 9(a) illustrates the load-strain relationship for slabs with varying thicknesses. As the load increased, the reinforcing bars began to carry significant tensile stresses to resist cracking and maintain the integrity of the connection. As the slab thickness increased, the reinforcing bars were less engaged in the loading process, leading to lower strain levels. As the load approached failure, the bars yielded, which contributed to a plateau in the load-strain curve and indicated

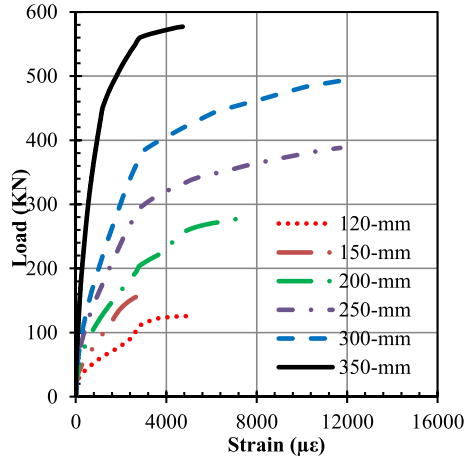
the redistribution of stresses between the concrete and the reinforcement. The yielding of the reinforcing bars marked a critical point in the structural response, signaling a shift to ductile behavior. Increasing the slab thickness from 120 mm to 150 mm, 200 mm, 250 mm, 300 mm, and 350 mm reduced the strains at the same load level by 53 %, 73 %, 86 %, 90 %, and 96 %, respectively. This significant reduction highlights the role of slab thickness in decreasing the demand on the reinforcing bars, thereby improving their performance.

The shear force (V), unbalanced moment (M_y), and shear stresses (V_c) are summarized in Table 3. These shear stresses were computed at a distance equal to half the average effective depth ($0.5d$) from the column face, as illustrated in Fig. 10, using Eqs. (1) and (2) prescribed by North American standards [17–18].

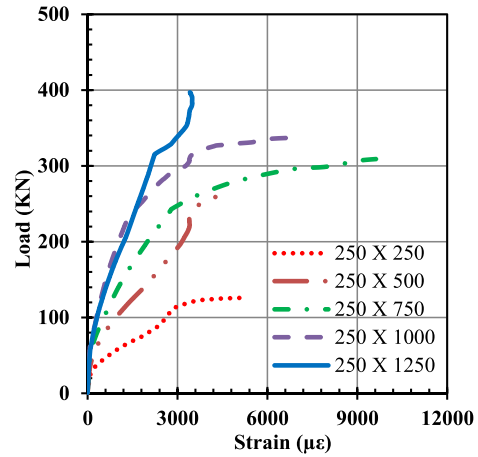
$$V_c = \frac{V}{d b_o} + \frac{M_y \gamma_V C_{AB}}{I_{Cy}} \quad (1)$$

$$\gamma_V = 1 - \frac{1}{1 + \frac{2}{3} \sqrt{\frac{b_1}{b_2}}} \quad (2)$$

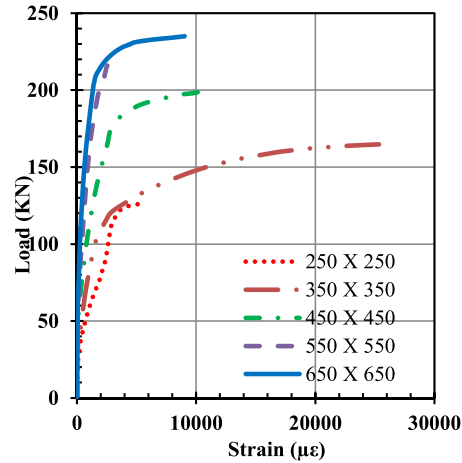
Where: b_o is the perimeter of the critical section for shear, b_1 and b_2 are the dimensions of the critical section in the span and perpendicular directions, respectively. C_{AB} is the distance from the centroid



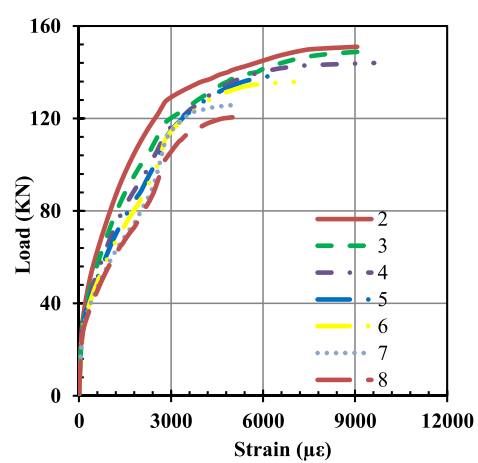
(a) Change in the slab-thickness



(b) Change in the column-aspect-ratio



(c) Change in the square column-stub



(d) Change in the span-to-depth ratio

Fig. 9. Load-strain curves for the studied parameters.

Table 3
Results of the model.

Connection's ID	Shear force V_{fe} (kN)	Unbalanced moment (kN.m)	Stresses at $0.5d$ from the column-face, V_C (MPa)	Deflection at $0.5d$ from the column-face (mm)	Strains in bars at $0.5d$ from the column-face (MPa)
Parameter number one: Slab thickness					
120-1-250-7	126	37.80	2.70	17.3	5190
150-1-250-7	175	52.50	2.48	12.1	3240
200-1-250-7	279	83.70	2.31	10.2	7640
250-1-250-7	388	116.4	2.10	8.33	11,700
300-1-250-7	493	147.9	1.88	7.18	11,600
350-1-250-7	577	173.1	1.63	5.56	4730
Parameter number two: Column aspect ratio					
120-1-250-7	126	37.80	2.70	17.3	5190
120-2-250-7	261	78.30	3.28	28.1	4690
120-3-250-7	309	92.70	2.69	32.9	9620
120-4-250-7	337	101.1	2.22	28.2	6780
120-5-250-7	397	119.1	2.10	40.7	3480
Parameter number three: Square column stub					
120-1-250-7	126	37.80	2.70	17.3	5190
120-1-350-7	165	49.50	2.34	27.0	26,000
120-1-450-7	199	59.70	2.07	39.8	10,300
120-1-550-7	216	64.80	1.74	46.5	2170
120-1-650-7	235	70.50	1.54	38.0	9030
Parameter number four: Span to depth ratio					
120-1-250-2	151	45.30	3.24	2.42	9070
120-1-250-3	149	44.70	3.20	3.93	9270
120-1-250-4	144	43.20	3.09	5.74	9620
120-1-250-5	139	41.70	2.98	7.16	6600
120-1-250-6	136	40.80	2.92	10.4	7260
120-1-250-7	126	37.80	2.70	17.3	5190
120-1-250-8	121	36.30	2.60	21.1	5360

of the critical section to the point of interest, and I_{cy} is the moment of inertia of the critical section about the y-axis.

Increasing the slab thickness from 150 mm to 200 mm, 250 mm, 300 mm, and 350 mm reduced punching shear stresses by 8 %, 14 %, 22 %, 30 %, and 40 %, respectively, compared to the 120-1-250-7 slab configuration. This reduction is due to the size effect, where greater slab thickness distributes shear forces and unbalanced moments over a larger area, thereby reducing shear stress intensity at the slab-column interface. As slab thickness increases, it becomes more effective at resisting punching shear.

Parameter 2: Column aspect ratio

To examine the impact of the column aspect ratio on the strength of the connections, five prototypes were constructed with column dimensions of 250×250 mm, 500×250 mm, 750×250 mm, $1000 \times$

250 mm, and 1250×250 mm, resulting in aspect ratios of 1, 2, 3, 4, and 5, respectively.

The load-deflection behavior is presented in Fig. 8(b), where the curvature of the slabs aligned with observations made in the previous section. As the column aspect ratio increased from 1 to 2, 3, 4, and 5, the deflection at the same shear level decreased by 67 %, 72 %, 74 %, and 78 %, respectively. This reduction in deflection was attributed to the increased stiffness of the slab-column connection as the column aspect ratio grew. A larger aspect ratio, driven by an increase in the greater dimension of the column cross-section, enhanced the overall stability and load distribution capabilities of the connection, thereby reducing deflection under applied shear forces.

However, at the ultimate failure stage, a significant increase in deflection was observed as the aspect ratio continued to grow. This behavior can be explained by the fact that as the aspect ratio increased, the column became more slender, making it more susceptible to instability or buckling under high loading conditions. As the greater dimension of the column cross-section increased, the column's lateral stiffness decreased, resulting in higher deformations at failure. This shift in behavior highlighted the complex interaction between column dimensions and slab deflection, showing that while increased column aspect ratios improved performance up to a point, they ultimately led to greater deflection at failure.

A similar trend was observed in the load-strain relationship, as shown in Fig. 9(b). As the column aspect ratio increased, the strains measured in the reinforcement at the column face decreased at the same shear level. Specifically, the strains were reduced by 64 %, 79 %, 88 %, and 89 % as the aspect ratio increased from 1 to 2, 3, 4, and 5, respectively. This reduction in strain can be attributed to the enhanced load distribution and stiffness provided by the increased column aspect ratio, which alleviated stress concentrations at the slab-column interface. The larger aspect ratio improved structural performance by reducing strain in the reinforcement, thereby contributing to a more efficient load transfer mechanism. At the failure stage, the reinforcement bars reached and surpassed the yielding point.

The punching shear strength increased by 21 %, from 2.7 MPa to 3.28 MPa (Table 3), when the column aspect ratio was increased from 1 to 2. This increase was attributed to the enhanced load distribution provided by the higher aspect ratio, which increased the stiffness of the slab-column connection and improved its ability to resist shear. A larger column aspect ratio resulted in a more stable connection by allowing a more efficient transfer of shear forces through the slab-column interface, thus increasing punching shear strength. However, beyond an aspect ratio of 2, the punching shear strength began to decrease. Specifically, the strength dropped by 18 %, 32 %, and 36 %, from 3.28 MPa to 2.69 MPa, 2.22 MPa, and 2.1 MPa, as the aspect ratio increased from 2 to 3, 4, and 5, respectively.

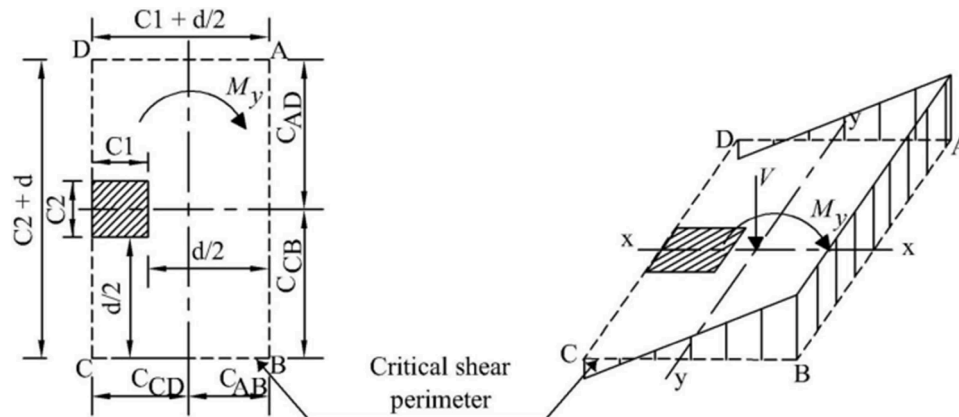


Fig. 10. Punching shear stresses.

This reduction in punching shear strength at higher aspect ratios was attributed to the increased slenderness of the column. As the column aspect ratio grew, the column became more prone to buckling under shear, negatively affecting the overall behavior of the slab-column connection. The higher aspect ratios also led to increased deflections and a reduction in the effectiveness of the reinforcement, both of which contributed to a decrease in punching shear capacity. Thus, while an increase in the aspect ratio initially enhanced the connection's shear strength, excessively high aspect ratios led to diminished performance due to instability and increased deformations.

The findings in the literature support similar trends observed in this study [19]. In one experiment, three edge slab-column connections were tested, each with different column aspect ratios of 1.0, 1.77, and 2.5. The experimental results indicated that increasing the column aspect ratio from 1.0 to 1.77 led to a noticeable increase in shear strength, rising from 3.23 MPa to 3.35 MPa. However, when the aspect ratio was further increased from 1.77 to 2.5, there was a slight reduction in punching shear strength, decreasing from 3.35 MPa to 3.33 MPa. These results suggest that while an increase in aspect ratio up to a certain point may enhance shear strength by improving load distribution and stability, further increases beyond an optimal value may lead to diminishing returns in punching shear strength.

• Parameter 3: Square column stub

To examine the effect of column size on the structural behavior of slab-column connections, five connections were constructed with columns having square cross sections. The side length of the square columns ranged from 250 mm to 650 mm, increasing incrementally by 100 mm.

The load-deflection curves for these connections, shown in Fig. 8(C), followed the same general trend described earlier. As the side length of the square column increased from 250 mm to 350 mm, 450 mm, 550 mm, and 650 mm, the punching shear strength progressively decreased by 25 %, 34 %, 41 %, and 50 %, respectively. As the column dimensions increased, the critical shear perimeter expanded, leading to a redistribution of stresses over a larger area.

Fig. 9(C) presents the load-strain curves for the analyzed connections. As the column dimensions increased, the strains in the reinforcement at the same load level significantly decreased. Specifically, the reduction in strain was approximately 42 %, 70 %, 85 %, and 91 % as the column side length increased from 250 mm to 350 mm, 450 mm, 550 mm, and 650 mm, respectively. This behavior can be attributed to the expanded load distribution area provided by larger column dimensions, which reduces the stress demand on the reinforcement near the column face, thereby lowering the strain levels in the bars.

The shear strength values presented in Table 3 reveal a noticeable reduction as the column dimensions increased. The punching shear strength dropped by 13 %, 23 %, 36 %, and 43 % as the column side length grew from 250 mm to 350 mm, 450 mm, 550 mm, and 650 mm, respectively. This reduction can be attributed to the increase in the critical shear perimeter, which reduced the stress concentration near the column face.

• Parameter 4: Span to depth ratio

Seven connections were designed and analyzed to assess the impact of the span-to-depth ratio on the structural behavior. For consistency, the slab thickness was maintained constant at 120 mm, aligning with the two parameters mentioned earlier. The column dimensions were also kept uniform across all connections, with a square cross-section measuring 250 mm on each side. To achieve the target span-to-depth ratios of 2, 3, 4, 5, 6, 7, and 8, the overall dimensions of the slabs were adjusted accordingly. It is important to

note that the span was defined as the distance from the column face to the support face, ensuring an accurate representation of the effective span in all configurations. Notably, this parameter did not receive attention in the existing literature.

The deflection behavior of the slabs is illustrated in Fig. 8(d), showcasing the pronounced influence of the span-to-depth ratio on the slab curvature. This parameter exhibited a significantly greater impact on deflection compared to the previously discussed parameters. Decreasing the span-to-depth ratio from 8 to 7, 6, 5, 4, 3, and 2 resulted in reductions in deflection at the same load level by approximately 26 %, 62 %, 74 %, 82 %, 88 %, and 93 %, respectively. The observed trend can be attributed to the fundamental relationship between span length and slab stiffness. As the span-to-depth ratio decreased, the effective span became shorter relative to the slab thickness, which significantly increased the structural stiffness of the slab-column system. This enhanced stiffness reduced the flexural deformations under applied loads. Moreover, smaller spans provided a more efficient load transfer mechanism, minimizing bending moments and improving overall resistance to deflection. Consequently, reducing the span-to-depth ratio led to substantial improvements in the load-deflection response, highlighting the critical role of geometric proportions in structural performance.

The strain behavior for the connections, as illustrated in Fig. 9(d), demonstrated a more modest sensitivity to changes in the span-to-depth ratio compared to its pronounced influence on deflection. The strain levels in the reinforcement exhibited only small reductions as the ratio decreased. Specifically, reducing the span-to-depth ratio from 8 to 7, 6, 5, 4, 3, and 2 resulted in reductions in strain at the same shear level by 5 %, 7 %, 10 %, 16 %, 27 %, and 41 %, respectively.

The variation in the punching shear strength, as presented in Table 3, demonstrated a clear trend in response to changes in the span-to-depth ratio. Reducing the span-to-depth ratio from 8 to 7, 6, 5, 4, 3, and 2 led to increases in punching shear strength by 4 %, 12 %, 15 %, 19 %, 23 %, and 25 %, respectively. This enhancement in the punching shear strength can be explained by the increased rigidity and stiffness of the slab as the span-to-depth ratio decreased. A smaller span-to-depth ratio resulted in a more restrained connection, which effectively distributed shear forces more evenly across the slab-column interface. With a reduced span, the slab would not experience large deformations, allowing the reinforcement to work more efficiently in carrying the applied shear forces. As the connection became stiffer and more resistant to bending, the overall punching shear strength improved.

4. Punching shear strength calculations according to various standards

The following sections outline the standards utilized to calculate the punching shear capacity of the finite element (FE) model.

4.1. American concrete institute standard, ACI 318–19 (22) [17]

Formula (3) was developed to determine the allowable punching shear strength. However, the joint ACI-ASCE committee [20] found that Formula (3) tended to overestimate strength predictions when the column aspect ratio exceeded two. This led to the establishment of Formula (4) as a refinement. Formula (5) was introduced to account for the impact of the ratio between the perimeter and effective depth on the shear capacity. The minimum value from these three formulas is recommended for determining the allowable punching shear capacity.

$$v_c = 0.33 \lambda \phi_c \sqrt{f'_c} \text{ (MPa)} \quad (3)$$

$$v_c = 0.083 \lambda \varphi_c \left(1 + \frac{4}{\beta_c}\right) \sqrt{f'_c} \text{ (MPa)} \quad (4)$$

$$v_c = 0.083 \lambda \varphi_c \left(2.0 + \alpha_s \frac{d}{b_0}\right) \sqrt{f'_c} \text{ (MPa)} \quad (5)$$

The value of α_s is 30 for edge connections.

4.2. Canadian standard (CSA-A23.3-2024) [18]

The Canadian standard adopts a comparable methodology to that of the American standard. It incorporates three equations (Eqs. (6)–8) designed to calculate the allowable punching shear strength.

$$v_c = 0.38 \lambda \varphi_c \sqrt{f'_c} \text{ (MPa)} \quad (6)$$

$$v_c = 0.19 \lambda \varphi_c \left(1 + \frac{2}{\beta_c}\right) \sqrt{f'_c} \text{ (MPa)} \quad (7)$$

$$v_c = \lambda \varphi_c \left(0.19 + \alpha_s \frac{d}{b_0}\right) \sqrt{f'_c} \text{ (MPa)} \quad (8)$$

The value of α_s is 3 for edge connections.

As outlined by the specifications, the compressive strength of concrete cylinders, denoted as f'_c , should not exceed 64 MPa. Furthermore, if the depth d surpasses 300 mm, the equations provided above need to be adjusted according to the formula in Eq. (9).

$$\text{Factor to consider the size effect} \frac{1300}{(1000 + d)} \quad (9)$$

4.3. Japanese standard for civil engineering (JSCE-2007) [21]

The Japanese guidelines determine the allowable punching shear capacity using Eq. (10), as outlined in Section 9.2.2.3. These guidelines consider factors such as the slab's average effective depth (d), the reinforcement ratio (ρ), the compressive strength of concrete (f'_c), and the perimeter of the column base (u).

$$v_c = \beta_d \beta_p \beta_r f_{pcd} / \gamma_b \quad (10)$$

Parameter to account for size effect $\beta_d = \sqrt[4]{\frac{1000}{d}} \leq 1.5$

Parameter to account for the reinforcement ratio $\beta_p = \sqrt[3]{100\rho} \leq 1.5$

Parameter to account for the applied load region $\beta_r = 1 + \frac{1}{1 + 0.25 \frac{u}{d}}$

Parameter to account for the concrete compressive strength $f_{pcd} =$

$$0.2 \sqrt{f'_c} \leq 1.2$$

4.4. Forecasts from the standards

The predictions provided by the three previously discussed standards are illustrated in Fig. 11 and Table 4, along with a comparison to the measured shear strength at the connections' failure.

• Parameter 1: Slab thickness

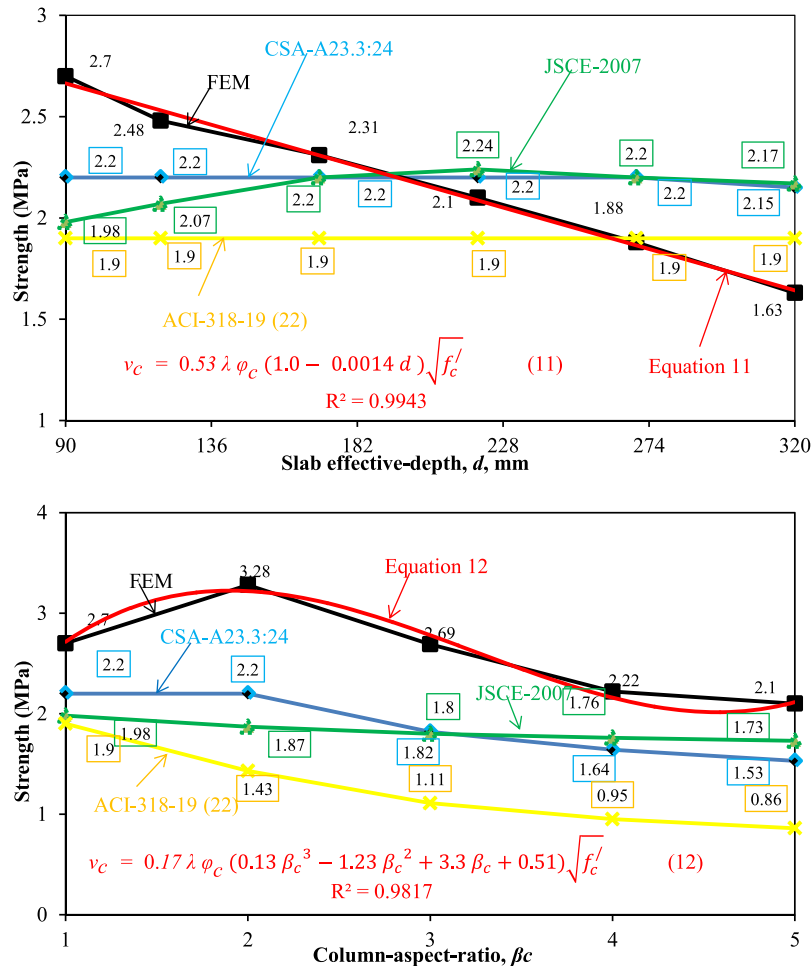


Fig. 11. Shear strength of the studied parameters.

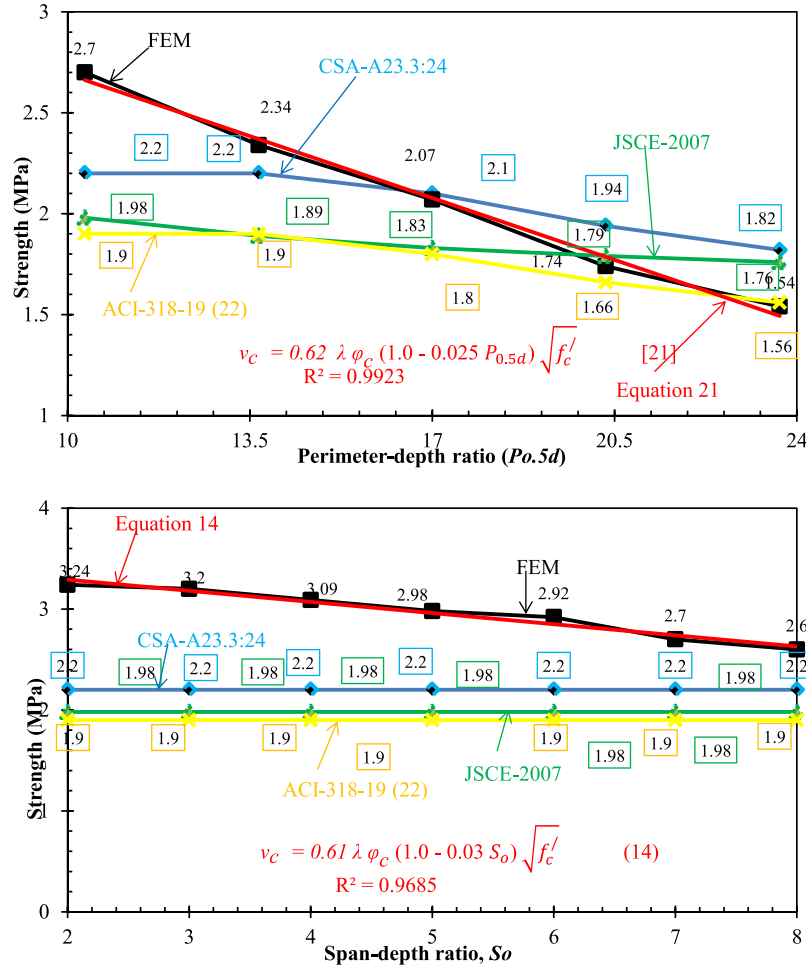


Fig. 11. (continued).

The standards evaluated in this study demonstrated limitations in accurately predicting the shear strength of the connections. None of them accurately predicted the decrease in strength associated with increasing slab thickness. The ACI 318–19 (22) provided a constant prediction of 1.9 MPa regardless of slab thickness. Similarly, the CSA-A23.3:24 delivered a consistent prediction of 2.2 MPa for slab thicknesses up to 300 mm. However, for a slab thickness of 350 mm, the predicted strength slightly decreased to approximately 2.15 MPa, influenced by the application of Eq. (9). In contrast, the JSCE-2007 standard exhibited a variable trend, with predictions increasing from 1.98 MPa for a slab thickness of 120 mm to 2.24 MPa at 250 mm. Beyond this point, the predicted strength declined, reaching 2.17 MPa at a thickness of 350 mm. To address the inaccuracies in the strength predictions made by the evaluated standards, Eq. (11) is introduced:

$$v_c = 0.53 \lambda \phi_c (1.0 - 0.0014 d) \sqrt{f'_c} \quad (11)$$

This equation is specifically designed to account for the effect of slab thickness on strength, providing a more reliable representation of the actual behavior. As shown in Fig. 11, Eq. (11) achieves a high level of accuracy in capturing the observed variations in slab strength, effectively compensating for the limitations of the existing standards.

• **Parameter 2: Column aspect ratio**

Among the three standards, the predictions from ACI 318–19 (22) were the most inaccurate and conservative. The predicted shear strength showed a significant decline, dropping from 1.9 MPa to 0.86 MPa as the column aspect ratio increased from 1 to 5. The CSA-A23.3:24 exhibited a similar but slightly more accurate trend compared to ACI 318–19 (22). It maintained a constant shear strength of 2.2 MPa for column aspect ratios of 1 and 2, but the values then decreased, reaching 1.53 MPa at a column aspect ratio of 5. The JSCE-2007 predictions were not significantly different from the other two standards. The shear strength decreased more gradually, from 1.98 MPa to 1.73 MPa, as the column aspect ratio increased from 1 to 5. These trends highlighted the inability of all three standards to accurately capture the impact of column aspect ratio on shear strength.

Although the three standards failed to accurately predict the strength of the connections due to variations in the column aspect ratio, their predictions remain conservative, unlike the previous parameter (slab thickness), where inaccuracies were less cautious. From a safety perspective, these conservative estimates can still be utilized. However, to overcome their limitations and improve accuracy, Eq. (12) is introduced, as demonstrated in Fig. 11. This equation provides a more precise representation of the connection strength, addressing the shortcomings of the standards while maintaining a focus on safety.

$$v_c = 0.17 \lambda \phi_c (0.13 \beta_c^3 - 1.23 \beta_c^2 + 3.3 \beta_c + 0.51) \sqrt{f'_c} \quad (12)$$

Table 4
The Standards' forecast.

Connection's ID	ACI 318–19 (22) forecast		CSA-A23.3:24 forecast		JSCE (2007) forecast	
	ν_{ACI} (MPa)	V_c/ν_{ACI}	ν_{CSA} (MPa)	V_c/ν_{CSA}	$\nu_{JSCE-07}$ (MPa)	$V_c/\nu_{JSCE-07}$
Parameter number one: Slab thickness						
120–1–250–7	1.90	1.42	2.20	1.23	1.98	1.36
150–1–250–7	1.90	1.31	2.20	1.13	2.07	1.20
200–1–250–7	1.90	1.22	2.20	1.05	2.20	1.05
250–1–250–7	1.90	1.11	2.20	0.95	2.24	0.94
300–1–250–7	1.90	0.99	2.20	0.85	2.20	0.85
350–1–250–7	1.90	0.86	2.15	0.76	2.17	0.75
Parameter number two: Column aspect ratio						
120–1–250–7	1.90	1.42	2.20	1.23	1.98	1.36
120–2–250–7	1.43	2.29	2.20	1.49	1.87	1.75
120–3–250–7	1.11	2.42	1.82	1.48	1.80	1.49
120–4–250–7	0.95	2.34	1.64	1.35	1.76	1.26
120–5–250–7	0.86	2.44	1.53	1.37	1.73	1.21
Parameter number three: Square column stub						
120–1–250–7	1.90	1.42	2.20	1.23	1.98	1.36
120–1–350–7	1.90	1.23	2.20	1.06	1.89	1.24
120–1–450–7	1.80	1.15	2.10	0.99	1.83	1.13
120–1–550–7	1.66	1.05	1.94	0.90	1.79	0.97
120–1–650–7	1.56	0.99	1.82	0.85	1.76	0.88
Parameter number four: Span to depth ratio						
120–1–250–2	1.90	1.71	2.20	1.47	1.98	1.64
120–1–250–3	1.90	1.68	2.20	1.45	1.98	1.62
120–1–250–4	1.90	1.63	2.20	1.40	1.98	1.56
120–1–250–5	1.90	1.57	2.20	1.35	1.98	1.51
120–1–250–6	1.90	1.54	2.20	1.33	1.98	1.47
120–1–250–7	1.90	1.42	2.20	1.23	1.98	1.36
120–1–250–8	1.90	1.37	2.20	1.18	1.98	1.31

• Parameter 3: Square column stub

Once again, the standards failed to accurately predict the reduction in the punching shear strength. The ACI 318–19 (22) generally provided conservative estimates. The code predicted a flat punching shear strength of 1.9 MPa for square columns with side lengths increasing from 250 mm to 350 mm. Beyond this range, the predicted shear strength decreased progressively to 1.56 MPa as the column side length reached 650 mm. Notably, at this endpoint, the code's prediction aligned closely with the actual result, differing by only 1 %. From a safety perspective, the code's conservative approach is commendable, but its accuracy remains limited. The CSA-A23.3:24 standard similarly provided a constant estimate of 2.2 MPa for column side lengths up to 350 mm. For side lengths between 350 mm and 650 mm, the predicted strength gradually reduced to 1.82 MPa. The JSCE-2007 standard presented a continuous decrease in the punching shear strength, from 1.98 MPa for a column with a 250 mm side length to 1.76 MPa for the one with a 650 mm side length.

Due to the inaccuracies observed in these standards, Eq. (13) is proposed as an alternative. As shown in Fig. 11, this equation offers significantly improved accuracy. Its primary variable is the perimeter around the column, calculated at half the average effective depth, relative to the slab's effective depth, denoted as $P_{0.5d}$.

$$v_c = 0.62 \lambda \varphi_c (1.0 - 0.025 P_{0.5d}) \sqrt{f'_c} \quad (13)$$

• Parameter 4: Span to depth ratio

The predictions provided by all the standards for the span-to-depth ratio were highly inaccurate. These standards relied on flat estimates and failed to account for the reduction in strength resulting from any increase in the span-to-depth ratio. Specifically, the American standard maintained a constant estimate of 1.9 MPa, the

Canadian standard predicted 2.2 MPa, and the Japanese standard provided a flat estimate of 1.98 MPa, regardless of any variation in the ratio.

Due to the inaccuracies in these predictions, Eq. (14) is proposed as a more reliable alternative. This equation accurately captures the influence of the span-to-depth ratio (S_o) on the actual strength, as demonstrated by its superior alignment with numerical results.

$$v_c = 0.61 \lambda \varphi_c (1.0 - 0.03 S_o) \sqrt{f'_c} \quad (14)$$

4.5. Evaluating the accuracy of proposed equations

To thoroughly assess the accuracy of the proposed Eqs. (11 to 14), a dataset comprising 15 steel-RC edge-slab column connections was collected from existing literature, as detailed in Appendix I. The appendix provides a comprehensive comparison of the actual stresses to the predicted stresses based on the standards considered in the current article. Failure stresses calculated using the proposed equations were also included. The selection of the governing failure stress followed the approach employed by North American standards, where the minimum value among the four equations was chosen as the representative stress.

The ACI standard produced a mean actual-to-predicted stress ratio of 1.54, while the CSA standard gave 1.34. The Japanese standard performed better, with a ratio of 1.22, which was closest to the actual values. However, the proposed equations yielded the most accurate results, with a ratio of 1.17, demonstrating their superior reliability in predicting the shear strength for these connections.

5. Conclusions

This study investigated the punching shear strength of steel-RC edge slab-column connections. Four key geometric parameters—slab thickness, column aspect ratio, square column stub, and span-to-depth ratio—were analyzed. Four equations were proposed in the study,

with results primarily focusing on steel-RC edge slab-column connections, as emphasized in the research. The author compared the proposed equations with three widely used standards: ACI 318–19 (22), CSA-A23.3:24, and JSCE-2007. Additionally, these equations were validated against data from 15 steel-RC edge slab-column connections available in the literature (Appendix I). The comparison demonstrated that the proposed equations yielded significantly better results than these standards, with improved accuracy in predicting punching shear strength. Therefore, in terms of applicability, the equations can confidently be applied to edge steel-RC connections. However, further investigation is required to verify their accuracy for interior and corner connections, which falls outside the scope of the current research.

1. Increasing slab thickness from 120 mm to 350 mm significantly reduced deflection during service conditions by 52 % to 95 %, but with diminishing effects near failure. The punching shear strength decreased by 40 % as slab thickness increased due to the redistribution of the shear stresses over a larger critical perimeter.
2. As the column aspect ratio increased from 1 to 5, the punching shear strength initially rose by 21 % (from 2.7 MPa to 3.28 MPa) at an aspect ratio of 2, but then declined by 36 % (to 2.1 MPa) at an aspect ratio of 5. This behavior reflects the trade-off between improved load distribution and increased slenderness leading to instability.
3. Expanding the column side length from 250 mm to 650 mm led to a 43 % reduction in punching shear strength (from 2.7 MPa to 1.54 MPa). Larger columns increased the critical shear perimeter, reducing stress concentration near the column face.
4. Reducing the span-to-depth ratio from 8 to 2 increased punching shear strength by 25 % (from 2.6 MPa to 3.24 MPa) while improving deflection behavior, with reductions of up to 93 %. This indicates improved stiffness and load transfer in connections with lower span-to-depth ratios.
5. All the standards evaluated—ACI 318–19 (22), CSA-A23.3:24, and JSCE-2007—failed to account for the effects of parameter variations accurately. Predictions for punching shear strength deviated by up to 144 % in some cases, underscoring the need for refined models.
6. The study introduced four equations tailored to account for geometric parameters, achieving significantly higher accuracy compared to the existing standards. These equations provide a robust framework for designing safer and more efficient edge steel-RC slab-column connections.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.rineng.2025.104581](https://doi.org/10.1016/j.rineng.2025.104581).

Appendix

Appendix I

Appendix I

Available literature.

Slab	L1 (mm)	L2 (mm)	Column dim. (mm)	Total depth (mm)	Effective Depth (mm)	f_c' (MPa)	flexural reinforcement p_f (%)	M/V (mm)	Shear force V_{exp} (kN)	Shear stresses $V_{exp,0.5d}$
[19]										
EX-S1	1000	1200	300 × 300	120	100	35.6	1.10	300	167.0	2.49
EX-S2	1000	1200	225 × 225	120	100	31.7	1.10	300	155.0	3.24
EX-S3	1000	1200	150 × 150	120	100	31.0	1.10	300	95.00	3.07
EX-S4	1000	1200	265 × 150	120	100	31.0	1.10	300	150.0	3.35
EX-S5	1000	1200	150 × 375	120	100	31.0	1.10	300	172.0	3.33
[22]										

(continued on next page)

Notations' list

b_1	= the width of the critical shear perimeter measured in the weaker direction (usually parallel to the shorter span or weaker axis of the slab-column connection), mm
b_2	= the width of the critical shear perimeter measured in the stronger direction (typically parallel to the longer span or stronger axis of the slab-column connection), mm
b_o , $P_{0.5d}$	= shear perimeter, mm
C_{AB}	= the distance from the centroid of the critical section to the point of interest, mm
d	= effective depth of tensile reinforcement, mm
f_c'	= specified compressive strength of concrete, MPa
I_{cy}	= the moment of inertia of the critical section about the y-axis, mm ⁴
M_y	= unbalanced moment, N mm
S_o	= span-to-depth ratio
u	= perimeter of the column cross section, mm
V_c	= punching shear strength, N
V , V_{fe}	= shear force, N
γ_v	= portion of the moment transferred by shear
γ_b	= member safety factor
λ	= factor accounting for concrete density
ρ	= reinforcement ratio
α_s	= factor to account for the type of connection
φ_c	= resistance factor for concrete
β_c	= column aspect ratio
β_d	= factor to account for the size effect
β_p	= parameter to account for the reinforcement ratio
β_r	= parameter to account for the applied load region
f_{pcd}	= parameter to account for the concrete compressive strength

Data availability: All the data are presented in the article.

CRediT authorship contribution statement

Ahmed Gouda: Writing – review & editing, Writing – original draft, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

I confirm that there is no conflict of interest.

Appendix I (continued)

Slab	L1 (mm)	L2 (mm)	Column dim. (mm)	Total depth (mm)	Effective Depth (mm)	f_c' (MPa)	flexural reinforcement p_f (%)	M/V (mm)	Shear force V_{exp} (kN)	Shear stresses $V_{exp,0.5d}$
S1	1380	1900	250 × 250	150	120	43.2	0.65	429	140.9	2.49
S4	1380	1900	250 × 250	150	120	32.2	1.07	428	141.0	2.49
[23]										
Z – 1 V (1)	965	1829	178 × 178	152	121	27.3	1.23	368	122.3	2.78
Z – V (1)	965	1829	267 × 267	152	121	34.3	1.23	393	215.3	3.30
Z—V (2)	965	1829	267 × 267	152	121	40.5	1.65	379	246.9	3.69
Z—V (3)	965	1829	267 × 267	152	118	38.7	2.23	386	268.2	4.21
Z—V (5)	965	1829	267 × 267	152	121	35.2	1.23	0.00	279.3	1.59
Z—V (6)	965	1829	267 × 267	152	121	31.3	1.23	753	117.0	2.82
Z—VI (1)	965	1829	356 × 356	152	121	26.0	1.23	403	265.1	2.93
[24]										
D15	1143	1219	152 × 152	76.0	57.0	31.1	1.50	872	11.60	2.06
Slab	$V_{exp,0.5d}$	CSA-A23.3:24	ACI 318–19 (22)	JSCE-2007	Proposed equations				Failure shear strength (Vf)**	$V_{exp,0.5d} / V_f$
		$V_{exp,0.5d} / V_{CSA}$	$V_{exp,0.5d} / V_{ACI}$	$V_{exp,0.5d} / V_{JSCE}$	EQ11	EQ12	EQ13	EQ14		
[19]										
EX-S1	2.49	1.10	1.26	1.08	2.72	2.75	2.68	2.88	2.68	0.93
EX-S2	3.24	1.51	1.74	1.42	2.57	2.59	2.73	2.64	2.57	1.26
EX-S3	3.07	1.45	1.67	1.27	2.54	2.57	2.89	2.53	2.53	1.21
EX-S4	3.35	1.58	1.82	1.46	2.54	3.05	2.69	2.65	2.54	1.31
EX-S5	3.33	1.58	1.81	1.51	2.54	2.94	2.7	2.53	2.53	1.32
[22]										
S1	2.49	1.00	1.15	1.21	2.9	3.03	3.23	2.88	2.88	0.86
S4	2.49	1.15	1.33	1.08	2.5	2.61	2.79	2.48	2.48	1.00
[23]										
Z – 1 V (1)	2.78	1.40	1.62	1.18	2.30	2.41	2.72	2.53	2.30	1.21
Z – V (1)	3.30	1.49	1.71	1.34	2.58	2.70	2.85	2.88	2.58	1.28
Z—V (2)	3.69	1.52	1.76	1.32	2.80	2.93	3.10	3.13	2.80	1.32
Z—V (3)	4.21	1.78	2.05	1.37	2.75	2.87	3.01	3.04	2.75	1.53
Z—V (5)	1.59	0.71	0.81	0.64	2.61	2.73	2.89	2.92	2.61	0.61
Z—V (6)	2.82	1.33	1.52	1.19	2.46	2.58	2.72	2.75	2.46	1.15
Z—VI (1)	2.93	1.51	1.74	1.43	2.24	2.35	2.31	2.54	2.24	1.31
[24]										
D15	2.06	0.97	1.12	0.84	2.72	2.57	2.59	1.63	1.63	1.26
Mean	–	1.34	1.54	1.22	–	–	–	–	–	1.17
SD	–	0.29	0.34	0.24	–	–	–	–	–	0.23
COV (%)	–	21.7	21.8	19.5	–	–	–	–	–	19.6

** The predicted shear capacity (Vf) represents the minimum value derived from the suggested formulas (11, 12, 13 and 14).

Data availability

Data will be made available on request.

References

[1] R.C. Elstner, E. Hognestad, Shearing strength of reinforced concrete slabs, *ACI Struct. J.* 53 (7) (1956) 29–58, <https://doi.org/10.14359/11501>.

[2] J. Moe, *Shearing Strength of Reinforced Concrete Slabs and Footings Under Concentrated Loads*, 19, Portland Cement Association, Research and Development Laboratories, 1961, pp. 1–30.

[3] S.H. Megally, A. Ghali, Punching shear design of earthquake-resistant slab-column connections, *ACI Struct. J.* 97 (5) (2000) 720–730, <https://doi.org/10.14359/8807>.

[4] S.H. Simmonds, S.D.B. Alexander, Truss model for edge column-slab connections, *ACI Struct. J.* 84 (4) (1987) 296–303, <https://doi.org/10.14359/1658>.

[5] E.F. El-Salakawy, M.A. Polak, M.H. Soliman, Reinforced concrete slab-column edge connections with shear studs, *Canad. J. Civil Eng.* 27 (2) (2000), <https://doi.org/10.1139/199-062>.

[6] A. Gouda, A.H. Ali, H.M. Mohamed, B. Benmokrane, Analysis of circular concrete members reinforced with composite glass-FRP spirals, *Compos. Struct.* 297 (2022) 115921, <https://doi.org/10.1016/j.compstruct.2022.115921>.

[7] M.A. Akinpelu, D.S. Gabriel, A.M. Salman, L.A. Raheem, Numerical study on the effect of different column shapes on punching shear behavior of flat slabs, *Results. Eng.* 19 (2023) 101345, <https://doi.org/10.1016/j.rineng.2023.101345>.

[8] Debiao Meng, Hengfei Yang, Shiyuan Yang, Yuting Zhang, Abílio M.P.De Jesus, José Correia, Tiago Fazerres-Ferradosa, Wojciech Macek, Ricardo Branco, Shun-Peng Zhu, Kriging-assisted hybrid reliability design and optimization of offshore wind turbine support structure based on a portfolio allocation strategy, *Ocean Engineering* 295 (2024) 116842, <https://doi.org/10.1016/j.oceaneng.2024.116842>.

[9] Debiao Meng, Shiyuan Yang, M.P. Abílio, Jesus de, Shun-Peng Zhu, A novel Kriging-model-assisted reliability-based multidisciplinary design optimization strategy and its application in the offshore wind turbine tower, *Renew. Energy* 203 (2023) 407–420, <https://doi.org/10.1016/j.renene.2022.12.062>.

[10] Debiao Meng, Shiyuan Yang, Hengfei Yang, Abílio M.P.De Jesus, José Correia, Shun-Peng Zhu, Intelligent-inspired framework for fatigue reliability evaluation of offshore wind turbine support structures under hybrid uncertainty, *Ocean Eng.* 307 (2024) 118213, <https://doi.org/10.1016/j.oceaneng.2024.118213>.

[11] A.H. Ali, A. Gouda, H.M. Mohamed, H.M. Esmael, Experimental and numerical analysis of steel and fiber-reinforced polymer concrete beams under transverse load, *ACI Struct. J.* 119 (4) (2022) 109–121, <https://doi.org/10.14359/51734651>.

[12] B.R. Alnemrawi, R.Z. Al-Rousan, Numerical study on the effect of different column shapes on punching shear behavior of flat slabs, *Results. Eng.* 24 (2024) 103315, <https://doi.org/10.1016/j.rineng.2024.103315>.

[13] E.F. El-Salakawy, M.A. Polak, M.H. Soliman, Reinforced concrete slab-column edge connections with openings, *ACI Struct. J.* 96 (1) (1999) 79–87, <https://doi.org/10.14359/598>.

[14] V. Cervenka, L. Jendele, J. Cervenka, *ATENA Program Documentation Part 1: Theory*, Cervenka Consulting Ltd, Prague, Czech Republic, 2023.

[15] CEB-FIP, *CEB-FIP model code 1990: design code, comité euro-international du béton - fédération Internationale de la précontrainte*, Bull. d'Inform. (213/214) (1993).

[16] D.A. Hordijk, PhD thesis, Delft University of Technology, Netherlands, 1991.

[17] ACI318, *Building Code Requirements For Structural concrete:(ACI 318-19 (22)); and Commentary (ACI 318R-19 (22))*, American Concrete Institute, 2022.

[18] CAN/CSA-A23.3, *Design of Concrete Structures (CAN/CSA-A23.3:24)*, Canadian Standards Association, 2024.

[19] A.G. Sherif, M.B. Emara, A.H. Ibrahim, S.A. Magd, Effect of the column dimensions on the punching shear strength of edge column-slab connections, *ACI Struct. J.* 232 (2005) 175–192, <https://doi.org/10.14359/14942>.

[20] ASCE-ACI Committee 426, *The shear strength of reinforced concrete members – slabs*, in: *ASCE Journal Proceeding* 100, 1974, pp. 1543–1591.

[21] JSCE, *Standard Specifications For Concrete Structures*, Japan Society of Civil Engineer, 2007.

- [22] J.D. Mortin, PhD dissertation, University of Calgary, 1989.
- [23] E.R.F. Zaghlool, E.R. F, PhD dissertation, University of Calgary, 1971, <https://doi.org/10.11575/PRISM/13724>.
- [24] N.W. Hanson, J.M. Hanson, Shear and moment transfer between concrete slabs and columns, J., PCA Res. Develop. Lab. 10 (1) (1968) 2–16.